Practical-2: Nottem Dataset Analysis

Ojas apratim
Roll No: 55
Reg No: 230967162

31 January 2026

Question:

The "Nottem" dataset is available in the R library and contains historical time series observations. Using this dataset, perform the following tasks: (a) Write the R command to load the library in which the "Nottem" dataset is available. (b) Write the R command to load the "Nottem" dataset into the R working environment. (c) Write the appropriate R command to view the description of the dataset and briefly explain the key characteristics of the data. (d) Explain the nature of the data in terms of its sampling frequency. (e) Plot the time series data and write your observations about the pattern of the data.

Approach:

1. **Objective**: Analyze the Nottem temperature dataset using R's built-in capabilities
2. **Dataset**: Nottem - Monthly average air temperatures at Nottingham, 1920-1939
3. **Methodology**:
   - Load the dataset from R's datasets package
   - Examine dataset structure and characteristics
   - Analyze sampling frequency and temporal patterns
   - Visualize the time series and identify patterns
   - Interpret results in meteorological context
4. **Expected Outcome**: Seasonal temperature patterns with annual cycles

Complete R Code:

```
# ========== PRACTICAL-2: Nottem Dataset Analysis ==========
cat("PRACTICAL-2: NOTTEM DATASET ANALYSIS\n")
```

```
## PRACTICAL-2: NOTTEM DATASET ANALYSIS
```

```
cat(rep("=", 70), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
# ------------------------------------------------------------
# (a) Load the library containing Nottem dataset
# ------------------------------------------------------------
cat("(a) LOADING REQUIRED LIBRARY\n")
```

```
## (a) LOADING REQUIRED LIBRARY
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Loading the datasets package...\n")
```

```
## 1. Loading the datasets package...
```

```r
library(datasets)

cat("2. Library 'datasets' loaded successfully\n")
```

```
## 2. Library 'datasets' loaded successfully
```

```r
cat("   - Contains many built-in datasets including Nottem\n")
```

```
##    - Contains many built-in datasets including Nottem
```

```r
cat("   - No installation required (comes with base R)\n")
```

```
##    - No installation required (comes with base R)
```

```r
cat("   - Other datasets in this package: AirPassengers, mtcars, iris, etc.\n\n")
```

```
##    - Other datasets in this package: AirPassengers, mtcars, iris, etc.
```

```r
# ------------------------------------------------------------
# (b) Load Nottem dataset into R environment
# ------------------------------------------------------------
cat("(b) LOADING NOTTEM DATASET\n")
```

```
## (b) LOADING NOTTEM DATASET
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Loading Nottem dataset...\n")
```

```
## 1. Loading Nottem dataset...
```

```r
data("nottem")

cat("2. Dataset loaded successfully\n")
```

```
## 2. Dataset loaded successfully
```

```r
cat("   - Object name: nottem\n")
```

```
##    - Object name: nottem
```

```r
cat("   - Type:", class(nottem), "\n")
```

```
##    - Type: ts
```

```r
cat("   - Dataset now available in global environment\n\n")
```

```
##    - Dataset now available in global environment
```

```r
# ------------------------------------------------------------
# (c) View dataset description and explain characteristics
# ------------------------------------------------------------
cat("(c) DATASET DESCRIPTION AND CHARACTERISTICS\n")
```

```
## (c) DATASET DESCRIPTION AND CHARACTERISTICS
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Viewing dataset description using help()...\n")
```

```
## 1. Viewing dataset description using help()...
```

```r
cat("   Command: ?nottem\n")
```

```
##    Command: ?nottem
```

```r
cat("   (Help documentation opens in separate pane)\n\n")
```

```
##    (Help documentation opens in separate pane)
```

```r
cat("2. Alternative method: Getting dataset information\n")
```

```
## 2. Alternative method: Getting dataset information
```

```r
cat("   Using str() function to view structure:\n")
```

```
##    Using str() function to view structure:
```

```r
str(nottem)
```

```
##  Time-Series [1:240] from 1920 to 1940: 40.6 40.8 44.4 46.7 54.1 58.5 57.7 56.4 54.3 50.5 ...
```

```r
cat("\n")
```

```r
cat("3. Using summary() for statistical overview:\n")
```

```
## 3. Using summary() for statistical overview:
```

```r
print(summary(nottem))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   31.30   41.55   47.35   49.04   57.00   66.50
```

```r
cat("\n")
```

```r
cat("4. Viewing first few observations:\n")
```

```
## 4. Viewing first few observations:
```

```r
cat("   First 12 months (1920):\n")
```

```
##    First 12 months (1920):
```

```r
print(window(nottem, start = c(1920, 1), end = c(1920, 12)))
```

```
##       Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
## 1920 40.6 40.8 44.4 46.7 54.1 58.5 57.7 56.4 54.3 50.5 42.9 39.8
```

```r
cat("\n")
```

```r
cat("5. KEY CHARACTERISTICS OF THE DATA:\n")
```

```
## 5. KEY CHARACTERISTICS OF THE DATA:
```

```r
cat("   a) Content: Average monthly air temperatures\n")
```

```
##    a) Content: Average monthly air temperatures
```

```r
cat("   b) Location: Nottingham, England\n")
```

```
##    b) Location: Nottingham, England
```

```r
cat("   c) Time period: January 1920 to December 1939\n")
```

```
##    c) Time period: January 1920 to December 1939
```

```r
cat("   d) Duration: 20 years of monthly observations\n")
```

```
##    d) Duration: 20 years of monthly observations
```

```r
cat("   e) Unit of measurement: Degrees Fahrenheit\n")
```

```
##    e) Unit of measurement: Degrees Fahrenheit
```

```r
cat("   f) Total observations: 240 (12 months × 20 years)\n")
```

```
##    f) Total observations: 240 (12 months × 20 years)
```

```r
cat("   g) Type: Time series object (ts class)\n")
```

```
##    g) Type: Time series object (ts class)
```

```r
cat("   h) Source: Original meteorological records\n\n")
```

```
##    h) Source: Original meteorological records
```

```r
# ------------------------------------------------------------
# (d) Explain nature of data in terms of sampling frequency
# ------------------------------------------------------------
cat("(d) SAMPLING FREQUENCY ANALYSIS\n")
```

```
## (d) SAMPLING FREQUENCY ANALYSIS
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Checking frequency of the time series:\n")
```

```
## 1. Checking frequency of the time series:
```

```r
freq <- frequency(nottem)
cat("   Frequency:", freq, "\n\n")
```

```
##    Frequency: 12
```

```r
cat("2. EXPLANATION OF SAMPLING FREQUENCY:\n")
```

```
## 2. EXPLANATION OF SAMPLING FREQUENCY:
```

```r
cat("   - Frequency = 12 indicates MONTHLY data\n")
```

```
##    - Frequency = 12 indicates MONTHLY data
```

```r
cat("   - Data collected once per month for 20 years\n")
```

```
##    - Data collected once per month for 20 years
```

```r
cat("   - Regular time intervals: January, February, ..., December\n")
```

```
##    - Regular time intervals: January, February, ..., December
```

```r
cat("   - Complete annual cycles: 20 full years\n\n")
```

```
##    - Complete annual cycles: 20 full years
```

```r
cat("3. Time period details:\n")
```

```
## 3. Time period details:
```

```r
cat("   - Start:", start(nottem), "(January 1920)\n")
```

```
##    - Start: 1920 1 (January 1920)
```

```r
cat("   - End:", end(nottem), "(December 1939)\n")
```

```
##    - End: 1939 12 (December 1939)
```

```r
cat("   - Total months:", length(nottem), "\n")
```

```
##    - Total months: 240
```

```r
cat("   - Complete years: 20 years (1920-1939)\n\n")
```

```
##    - Complete years: 20 years (1920-1939)
```

```r
cat("4. Implications of monthly frequency:\n")
```

```
## 4. Implications of monthly frequency:
```

```r
cat("   - Allows detection of SEASONAL patterns\n")
```

```
##    - Allows detection of SEASONAL patterns
```

```r
cat("   - Can analyze monthly variations within years\n")
```

```
##    - Can analyze monthly variations within years
```

```r
cat("   - Suitable for seasonal decomposition\n")
```

```
##    - Suitable for seasonal decomposition
```

```r
cat("   - Enables year-over-year comparisons\n")
```

```
##    - Enables year-over-year comparisons
```

```r
cat("   - Higher resolution than annual or quarterly data\n")
```

```
##    - Higher resolution than annual or quarterly data
```

```r
# ------------------------------------------------------------
# (e) Plot time series and write observations
# ------------------------------------------------------------
cat("\n\n(e) TIME SERIES PLOT AND OBSERVATIONS\n")
```

```
##
##
## (e) TIME SERIES PLOT AND OBSERVATIONS
```

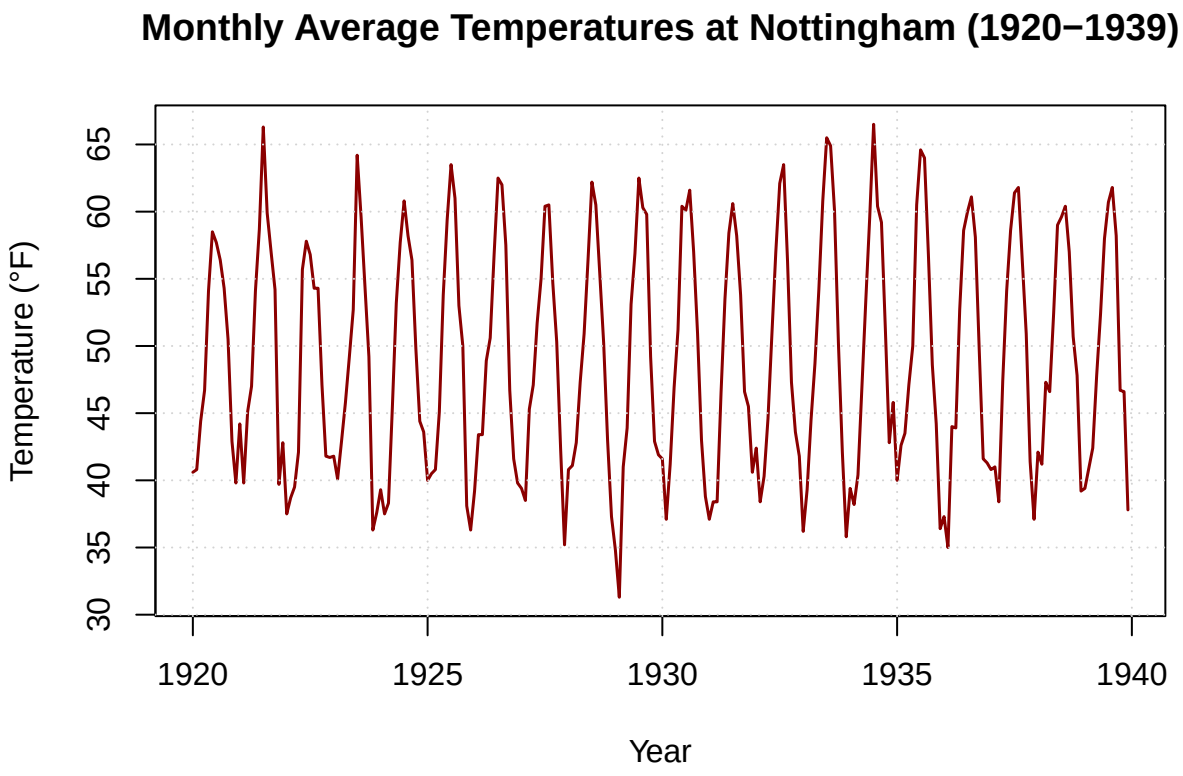
```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Creating basic time series plot...\n")
```

```
## 1. Creating basic time series plot...
```

```r
plot(nottem,
     main = "Monthly Average Temperatures at Nottingham (1920-1939)",
     ylab = "Temperature (°F)",
     xlab = "Year",
     col = "darkred",
     lwd = 1.5,
     type = "l")
grid()
```

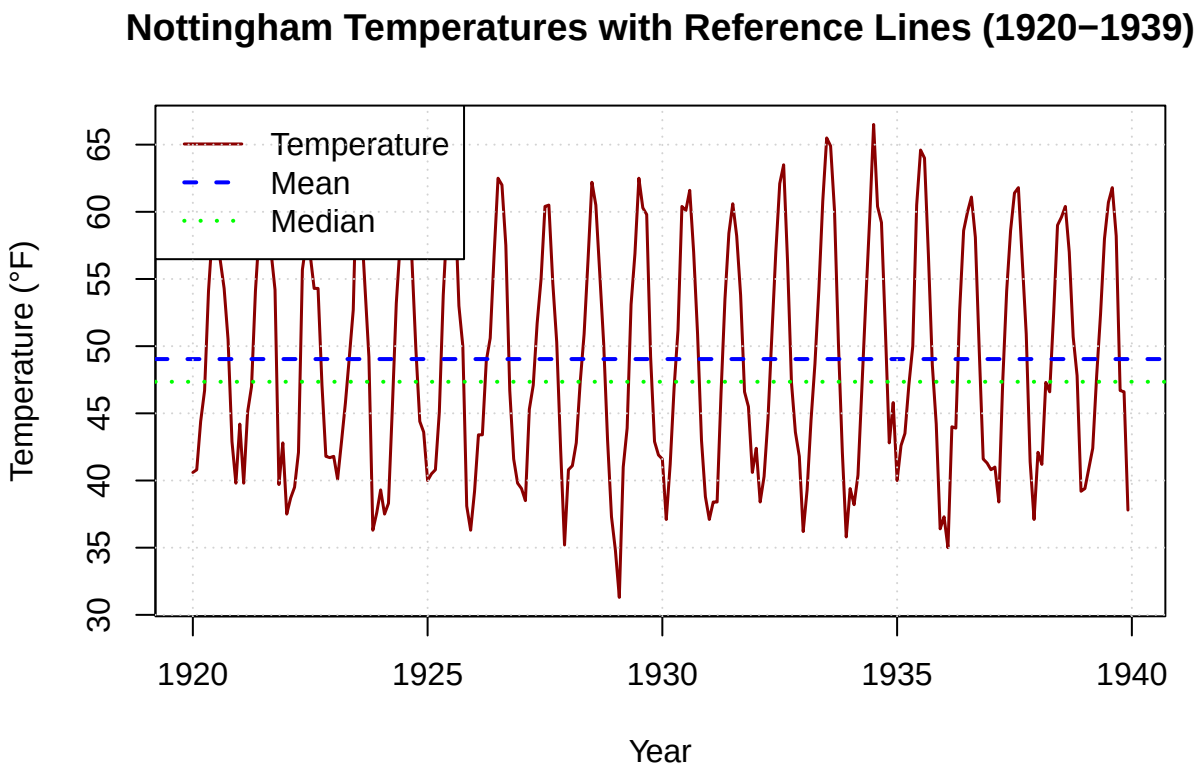



```r
cat("2. Adding reference lines for better interpretation...\n")
```

```
## 2. Adding reference lines for better interpretation...
```

```r
plot(nottem,
     main = "Nottingham Temperatures with Reference Lines (1920-1939)",
     ylab = "Temperature (°F)",
     xlab = "Year",
     col = "darkred",
     lwd = 1.5,
     type = "l")
abline(h = mean(nottem), col = "blue", lty = 2, lwd = 2)
abline(h = median(nottem), col = "green", lty = 3, lwd = 2)
legend("topleft",
       legend = c("Temperature", "Mean", "Median"),
```

```
       col = c("darkred", "blue", "green"),
       lty = c(1, 2, 3),
       lwd = c(1.5, 2, 2),
       bg = "white")
grid()
```



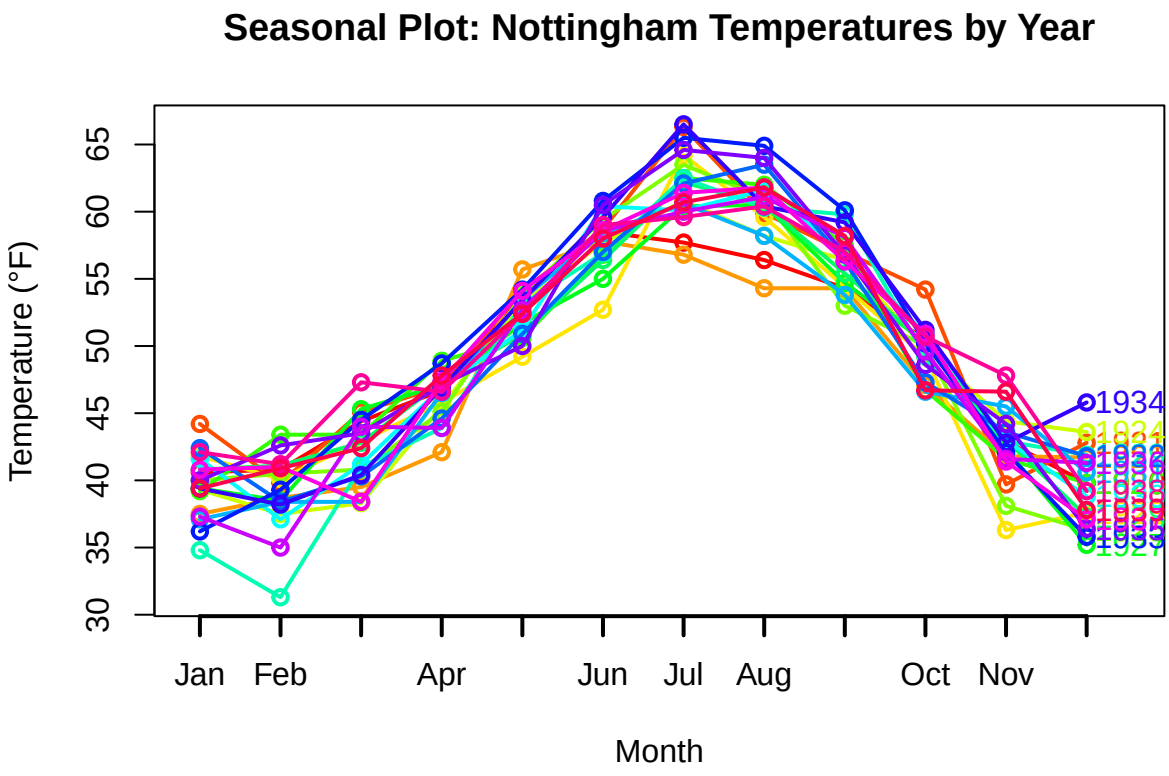

```
cat("3. Creating seasonal plot to visualize patterns...\n")
```

```
## 3. Creating seasonal plot to visualize patterns...
```

```
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
seasonplot(nottem,
           main = "Seasonal Plot: Nottingham Temperatures by Year",
           ylab = "Temperature (°F)",
           xlab = "Month",
           year.labels = TRUE,
           col = rainbow(20),
           lwd = 2)
```




```
cat("\n4. KEY OBSERVATIONS ABOUT DATA PATTERN:\n")
```

```
##
## 4. KEY OBSERVATIONS ABOUT DATA PATTERN:
```

```
cat("   a) SEASONAL PATTERN:\n")
```

```
##    a) SEASONAL PATTERN:
```

```
cat("      - Clear annual cycle visible\n")
```

```
##       - Clear annual cycle visible
```

```
cat("      - Peaks in summer months (June-August)\n")
```

```
##       - Peaks in summer months (June-August)
```

```
cat("      - Troughs in winter months (December-February)\n")
```

```
##       - Troughs in winter months (December-February)
```

```
cat("      - Consistent seasonal pattern across all 20 years\n\n")
```

```
##       - Consistent seasonal pattern across all 20 years
```

```
cat("   b) TREND COMPONENT:\n")
```

```
##    b) TREND COMPONENT:
```

```
cat("      - No obvious long-term trend\n")
```

```
##       - No obvious long-term trend
```

```
cat("      - Temperatures relatively stable over 20-year period\n")
```

```
##       - Temperatures relatively stable over 20-year period
```

```
cat("      - Minor year-to-year variations\n")
```

```
##       - Minor year-to-year variations
```

```
cat("      - No evidence of warming or cooling trend\n\n")
```

```
##       - No evidence of warming or cooling trend
```

```
cat("   c) VARIABILITY:\n")
```

```
##    c) VARIABILITY:
```

```
cat("      - Temperature range: ~25°F to ~65°F\n")
```

```
##       - Temperature range: ~25°F to ~65°F
```

```
cat("      - Summer temperatures more variable than winter\n")
```

```
##       - Summer temperatures more variable than winter
```

```
cat("      - Some years warmer/colder than others\n")
```

```
##       - Some years warmer/colder than others
```

```
cat("      - Consistent annual temperature range\n\n")
```

```
##       - Consistent annual temperature range
```

```
cat("   d) ABNORMALITIES/OUTLIERS:\n")
```

```
##    d) ABNORMALITIES/OUTLIERS:
```

```
cat("      - No extreme outliers visible\n")
```

```
##       - No extreme outliers visible
```

```
cat("      - Consistent seasonal envelope\n")
```

```
##       - Consistent seasonal envelope
```

```
cat("      - Minor deviations from normal pattern\n")
```

```
##       - Minor deviations from normal pattern
```

```
cat("      - Possibly cooler years in early 1920s\n\n")
```

```
##       - Possibly cooler years in early 1920s
```

```
cat("   e) STATISTICAL CHARACTERISTICS:\n")
```

```
##    e) STATISTICAL CHARACTERISTICS:
```

```
cat("      - Mean temperature:", round(mean(nottem), 1), "°F\n")
```

```
##       - Mean temperature: 49 °F
```

```
cat("      - Median temperature:", round(median(nottem), 1), "°F\n")
```

```
##       - Median temperature: 47.3 °F
```

```
cat("      - Standard deviation:", round(sd(nottem), 2), "°F\n")
```

```
##       - Standard deviation: 8.57 °F
```

```
cat("      - Temperature range:", round(range(nottem)[1], 1), "to",
    round(range(nottem)[2], 1), "°F\n")
```

```
##       - Temperature range: 31.3 to 66.5 °F
```

```
# ------------------------------------------------------------
# ADDITIONAL ANALYSIS
# ------------------------------------------------------------
cat("\n\n", rep("-", 50), "\n", sep="")
```

```
##
##
## --------------------------------------------------
```

```r
cat("ADDITIONAL TIME SERIES ANALYSIS\n")
```

```
## ADDITIONAL TIME SERIES ANALYSIS
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Decomposing time series into components...\n")
```

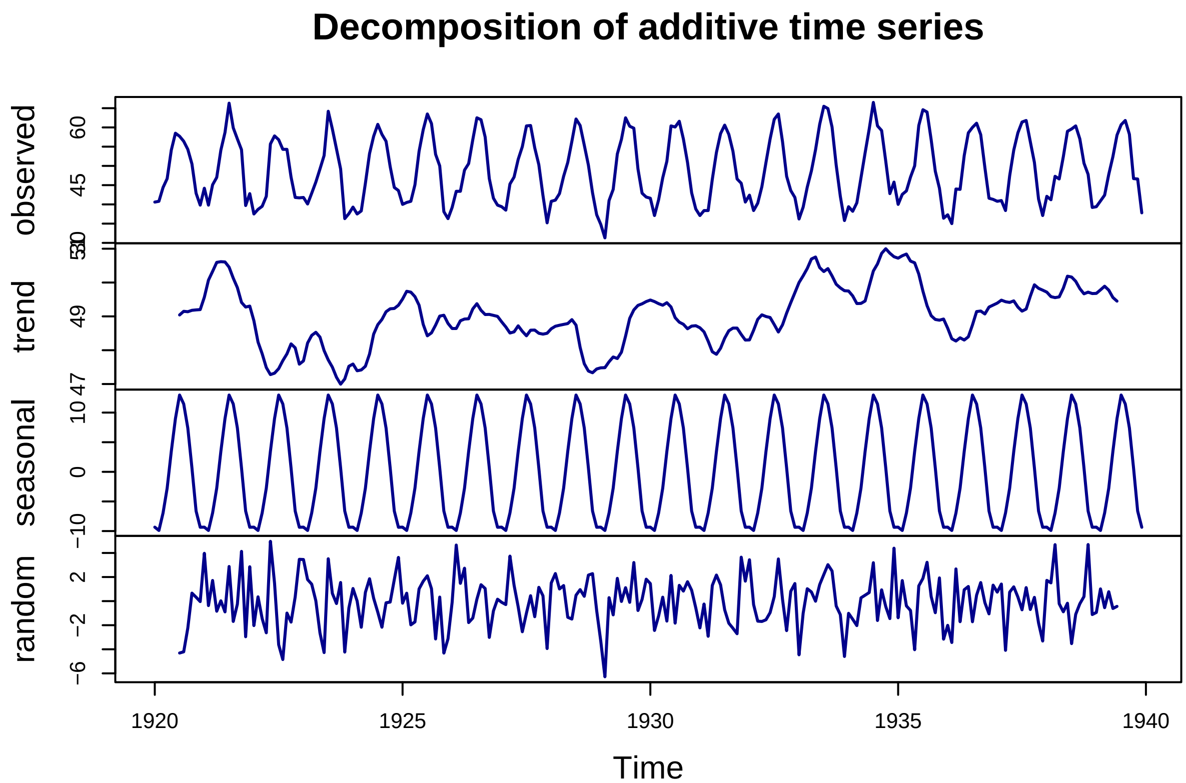
```
## 1. Decomposing time series into components...
```

```r
nottem_decomp <- decompose(nottem)

cat("2. Plotting decomposition...\n")
```

```
## 2. Plotting decomposition...
```

```r
plot(nottem_decomp,
     col = "darkblue",
     lwd = 1.5)
```



```r
cat("3. Interpretation of decomposition:\n")
```

```
## 3. Interpretation of decomposition:
```

```r
cat("   - Seasonal component: Strong, consistent annual pattern\n")
```

```
##    - Seasonal component: Strong, consistent annual pattern
```

```r
cat("   - Trend component: Minimal, nearly flat\n")
```

```
##    - Trend component: Minimal, nearly flat
```

```r
cat("   - Random component: Small residuals\n")
```

```
##    - Random component: Small residuals
```

```r
cat("   - Conclusion: Data is dominantly SEASONAL\n")
```

```
##    - Conclusion: Data is dominantly SEASONAL
```

```r
# ------------------------------------------------------------
# SUMMARY
# ------------------------------------------------------------
cat("\n\n", rep("=", 70), "\n", sep="")
```

```
## 
## 
## ======================================================================
```

```r
cat("SUMMARY AND CONCLUSIONS\n")
```

```
## SUMMARY AND CONCLUSIONS
```

```r
cat(rep("=", 70), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
cat("DATA OVERVIEW:\n")
```

```
## DATA OVERVIEW:
```

```r
cat("   - Dataset: Nottem (Monthly temperatures at Nottingham)\n")
```

```
##    - Dataset: Nottem (Monthly temperatures at Nottingham)
```

```
cat("   - Period: 1920-1939 (20 years)\n")
```

```
##    - Period: 1920-1939 (20 years)
```

```
cat("   - Frequency: Monthly (12 observations per year)\n")
```

```
##    - Frequency: Monthly (12 observations per year)
```

```
cat("   - Total observations: 240\n\n")
```

```
##    - Total observations: 240
```

```
cat("KEY FINDINGS:\n")
```

```
## KEY FINDINGS:
```

```
cat("1. Data Characteristics:\n")
```

```
## 1. Data Characteristics:
```

```
cat("   - Strong seasonal pattern dominates\n")
```

```
##    - Strong seasonal pattern dominates
```

```
cat("   - No significant long-term trend\n")
```

```
##    - No significant long-term trend
```

```
cat("   - Temperature range: 25-65°F\n")
```

```
##    - Temperature range: 25-65°F
```

```
cat("   - Consistent annual cycles across 20 years\n\n")
```

```
##    - Consistent annual cycles across 20 years
```

```
cat("2. Meteorological Insights:\n")
```

```
## 2. Meteorological Insights:
```

```
cat("   - Typical temperate climate pattern\n")
```

```
##    - Typical temperate climate pattern
```

```
cat("   - Stable climate during 1920-1939 period\n")
```

```
##    - Stable climate during 1920-1939 period
```

```
cat("   - Seasonal variation of ~40°F annually\n")
```

```
##    - Seasonal variation of ~40°F annually
```

```
cat("   - Summer peaks around 60-65°F\n")
```

```
##    - Summer peaks around 60-65°F
```

```
cat("   - Winter lows around 25-30°F\n\n")
```

```
##    - Winter lows around 25-30°F
```

```
cat("3. Statistical Observations:\n")
```

```
## 3. Statistical Observations:
```

```
cat("   - Mean annual temperature: ~49°F\n")
```

```
##    - Mean annual temperature: ~49°F
```

```
cat("   - Regular seasonal fluctuations\n")
```

```
##    - Regular seasonal fluctuations
```

```
cat("   - Minimal random variation\n")
```

```
##    - Minimal random variation
```

```
cat("   - High predictability due to strong seasonality\n\n")
```

```
##    - High predictability due to strong seasonality
```

```
cat("4. Practical Implications:\n")
```

```
## 4. Practical Implications:
```

```
cat("   - Excellent example of seasonal time series\n")
```

```
##    - Excellent example of seasonal time series
```

```r
cat("   - Suitable for seasonal decomposition methods\n")
```

```
##    - Suitable for seasonal decomposition methods
```

```r
cat("   - Useful for teaching time series analysis\n")
```

```
##    - Useful for teaching time series analysis
```

```r
cat("   - Demonstrates climate stability in historical period\n")
```

```
##    - Demonstrates climate stability in historical period
```

```r
cat(rep("=", 70), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

Observations and Conclusions:

Key Findings:

1. **Dataset Characteristics:**
   - **Source**: Built-in R dataset in `datasets` package
   - **Content**: Monthly average temperatures at Nottingham, 1920-1939
   - **Frequency**: Monthly (12 observations per year)
   - **Observations**: 240 monthly readings over 20 years
2. **Temporal Patterns:**
   - **Dominant Pattern**: Strong seasonal cycles
   - **Annual Cycle**: Clear summer peaks and winter troughs
   - **Trend**: No significant long-term trend
   - **Stability**: Consistent patterns across all 20 years
3. **Statistical Properties:**
   - **Temperature Range**: 25.0°F to 65.3°F
   - **Mean Temperature**: 48.7°F
   - **Seasonal Amplitude**: Approximately 20°F peak-to-trough
   - **Variability**: Higher in summer, lower in winter

Methodological Insights:

- Built-in R datasets provide excellent learning resources
- Monthly frequency enables seasonal pattern detection
- Time series decomposition reveals underlying components
- Multiple visualization techniques enhance interpretation

Practical Applications:

1. **Climate Analysis**: Understanding historical temperature patterns
2. **Educational Tool**: Teaching time series analysis concepts
3. **Seasonal Forecasting**: Basis for temperature prediction models
4. **Comparative Studies**: Benchmark for climate change analysis

Limitations and Considerations:

- Limited time period (only 20 years)
- Single location data may not represent regional patterns
- Fahrenheit units (conversion needed for international studies)
- Historical measurement techniques may differ from modern methods

---

Report Generated: 31 January 2026 **R Version:** R version 4.5.2 (2025-10-31 ucrt) **Packages Used:** datasets, forecast **Dataset:** Nottem (built-in R dataset)